

E-COMMERCE SALES ANALYTICS

Final Data Analytics Project Report

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1. PROJECT OVERVIEW

E-Commerce Sales Analytics is an end-to-end data analytics project developed to analyze sales performance, customer segments, product performance, regional performance, and shipping efficiency.

The project uses an e-commerce sales dataset and applies data preparation, exploratory data analysis, data visualization, and business intelligence techniques to transform raw transactional data into meaningful business insights.

Python-based data analysis was used for data preparation, exploration, and visualization, while Microsoft Power BI was used to develop an interactive dashboard containing key performance indicators, charts, and filters.

The final dashboard provides a consolidated view of important business metrics such as total sales, total orders, total customers, total products, regional sales, category performance, customer segment contribution, top-selling products, and average shipping time.

The project demonstrates the complete data analytics workflow from raw data to actionable business recommendations.

2. PROBLEM STATEMENT

E-commerce businesses generate large amounts of transactional data containing information about customers, products, sales, regions, categories, and shipping.

However, raw transactional data alone does not provide clear information about business performance.

The main problem addressed by this project is:

How can e-commerce sales data be analyzed and visualized to identify important sales trends, high-performing regions and products, customer segment contributions, and shipping patterns that can support better business decisions?

The project aims to convert raw e-commerce data into useful information through systematic data cleaning, exploratory analysis, visualization, and dashboard development.

3. PROJECT OBJECTIVES

The major objectives of the project are:

1. To understand and inspect the selected e-commerce dataset.
2. To identify the important columns and data types.
3. To detect and handle missing or incorrect data.
4. To identify and remove duplicate records where required.
5. To perform exploratory data analysis.
6. To identify sales trends and patterns.
7. To analyze sales performance by region and category.
8. To identify the highest-performing products.
9. To analyze customer segment contribution.
10. To analyze shipping performance by shipping mode.
11. To create meaningful data visualizations.
12. To develop an interactive Power BI dashboard.
13. To identify important business insights.
14. To provide data-driven business recommendations.

4. DATASET DESCRIPTION

The project uses a real-world **E-Commerce Sales Dataset** containing transactional information about customer orders, products, sales, regions, customer segments, and shipping.

After the data cleaning and preparation process, the final dataset contains **9,800 records and 24 columns**. The dataset covers orders placed between **2015 and 2018** and provides information required to analyze sales performance, product performance, customer segments, regional performance, and shipping efficiency.

The cleaned dataset was used for **Exploratory Data Analysis (EDA), data visualization, and Power BI dashboard development**.

4.1 Dataset Summary

Attribute	Details
Number of Records	9,800
Number of Columns	24
Time Period	2015–2018
Total Orders	4,922
Total Customers	793
Total Products	1,861
Product Categories	3
Product Sub-Categories	17
Regions	4
Customer Segments	3
Shipping Modes	4
Missing Values	None after cleaning
Duplicate Records	None after cleaning
Total Sales	Approximately 2.26M

4.2 Dataset Fields

The cleaned dataset contains the following 24 fields:

Field	Description
Row ID	Unique identifier for each row
Order ID	Unique identifier of the customer order
Order Date	Date on which the order was placed
Ship Date	Date on which the order was shipped
Ship Mode	Shipping method selected for the order
Customer ID	Unique identifier of the customer
Customer Name	Name of the customer
Segment	Customer segment
Country	Country associated with the order
City	City associated with the order
State	State associated with the order
Postal Code	Postal code of the customer location

Region	Geographical sales region
Product ID	Unique identifier of the product
Category	Main product category
Sub-Category	Product sub-category
Product Name	Name of the product
Sales	Revenue generated by the transaction
Year	Year extracted from the Order Date
Month	Numerical month extracted from the Order Date
Month_Name	Name of the month
Year_Month	Combined year and month used for time-based analysis
Quarter	Quarter extracted from the Order Date
Shipping_Days	Number of days between Order Date and Ship Date

4.3 Main Analytical Dimensions

The cleaned dataset supports analysis across several important business dimensions.

Sales

The `Sales` field represents the revenue generated from each transaction and is the primary measure used to evaluate business performance.

Orders

The dataset contains **4,922 unique orders**, providing a basis for analyzing order activity and overall sales performance.

Customers

The dataset contains **793 customers**, allowing customer-level and customer-segment analysis.

Products

The dataset contains **1,861 products**, allowing analysis of product-level sales performance and identification of the Top 10 products by sales.

Categories

The dataset contains three major product categories:

- Technology
- Furniture
- Office Supplies

Regions

Sales performance can be compared across four geographical regions:

- West
- East
- Central
- South

Customer Segments

Customers are divided into three segments:

- Consumer
- Corporate
- Home Office

Shipping Modes

The dataset contains four shipping modes:

- Standard Class
- Second Class
- First Class
- Same Day

4.4 Derived Analytical Fields

Several analytical fields were prepared to support time-based and operational analysis.

These include:

- **Year**
- **Month**
- **Month_Name**
- **Year_Month**
- **Quarter**
- **Shipping_Days**

The `Year` field supports yearly sales analysis, while `Month`, `Month_Name`, and `Year_Month` can be used for monthly trend analysis.

The `Quarter` field supports quarterly analysis.

The `Shipping_Days` field represents the difference between the order date and shipping date and is used to analyze average shipping time by shipping mode.

4.5 Data Quality After Cleaning

After the data preparation and cleaning process, the final dataset contains:

- **9,800 records**
- **24 columns**
- **No missing values**
- **No duplicate records**

The cleaned dataset was therefore used as the final source for exploratory data analysis, visualization, and Power BI dashboard development.

4.6 Dataset Suitability

The dataset is suitable for e-commerce business analysis because it contains information related to:

- Sales revenue
- Orders
- Customers
- Products
- Product categories
- Product sub-categories

- Regions
- Customer segments
- Shipping methods
- Order dates
- Shipping dates

This combination of information allows the project to analyze both **sales performance** and **operational performance**.

The dataset therefore provides a suitable foundation for identifying business trends, comparing performance across different dimensions, and developing data-driven recommendations.

5. TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used in the project:

Python

Python was used for:

- Data loading
- Data inspection
- Data cleaning
- Exploratory Data Analysis
- Statistical analysis
- Data visualization

Python Libraries

The major libraries used include:

- Pandas
- NumPy
- Matplotlib
- Seaborn

Microsoft Power BI

Power BI was used to:

- Import the prepared dataset
- Create calculated measures
- Create KPI cards
- Create interactive visualizations
- Add slicers
- Build the final business dashboard

Git

Git was used for:

- Version control
- Tracking project changes
- Creating commits
- Maintaining project history

GitHub

GitHub was used for:

- Hosting the project repository
- Maintaining project documentation
- Storing project files
- Creating a portfolio-ready project

6. DATA PREPARATION AND CLEANING

Data preparation and cleaning were performed to ensure that the e-commerce dataset was suitable for Exploratory Data Analysis, visualization, and Power BI dashboard development.

The original dataset was inspected, cleaned, and transformed into a structured analytical dataset containing **9,800 records and 24 columns**.

6.1 Data Loading

The e-commerce sales dataset was loaded into Python using the **Pandas** library.

The dataset was initially inspected to understand:

- Number of records
- Number of columns
- Column names
- Data types
- Missing values
- Duplicate records
- Numerical and categorical fields

Basic Pandas functions such as `head()`, `info()`, `describe()`, `isnull()`, and `duplicated()` were used during the inspection process.

6.2 Data Structure Inspection

The dataset was examined to identify the different types of information available.

The main types of fields included:

- Identifier fields such as Order ID, Customer ID, and Product ID
- Date fields such as Order Date and Ship Date
- Categorical fields such as Category, Region, Segment, and Ship Mode
- Numerical fields such as Sales and Postal Code

This inspection helped determine which columns could be used for different types of analysis.

6.3 Missing Value Handling

The dataset was checked for missing values across all columns.

Missing-value analysis was performed before starting the EDA process. Where missing or incomplete values were identified, they were handled during the cleaning process so that they would not negatively affect the analysis.

After the cleaning process, the final dataset contained:

No missing values.

Therefore, the cleaned dataset was suitable for further analysis.

6.4 Duplicate Record Handling

Duplicate records were checked to ensure that the same transaction was not counted more than once.

Duplicate rows were identified using duplicate-record checks.

After the cleaning process, the final dataset contained:

No duplicate records.

This helped prevent incorrect calculations and double counting during sales analysis.

6.5 Date Data Preparation

The `Order Date` and `Ship Date` fields were prepared as date fields.

Additional time-based fields were created to support analysis at different time levels.

The following analytical fields were prepared:

- `Year`
- `Month`
- `Month_Name`
- `Year_Month`
- `Quarter`

These fields allow sales to be analyzed by year, month, and quarter.

6.6 Shipping Time Calculation

A new field called `Shipping_Days` was prepared using the difference between the order date and shipping date.

The field represents the number of days required to ship an order.

This field was used in the Power BI dashboard to calculate and compare the average shipping duration for different shipping modes.

6.7 Data Type Validation

The data types of the columns were checked and prepared appropriately for analysis.

Date fields were treated as date values, numerical fields were used for calculations, and categorical fields were used for grouping and filtering.

This ensured that the dataset could be correctly analyzed in Python and Power BI.

6.8 Data Validation

After the cleaning process, the dataset was reviewed to ensure that the data was suitable for analysis.

The final cleaned dataset contained:

Data Quality Check	Result
Records	9,800
Columns	24
Missing Values	None
Duplicate Records	None
Date Fields	Prepared
Analytical Time Fields	Created
Shipping Days	Calculated
Dataset Status	Ready for Analysis

6.9 Final Prepared Dataset

After completing the data preparation and cleaning process, the dataset was ready for:

- Exploratory Data Analysis
- Descriptive statistics
- Correlation analysis
- Outlier analysis
- Data visualization
- Power BI dashboard development
- Business insight generation

The cleaned dataset was used consistently throughout the remaining stages of the project to ensure that the analysis and dashboard were based on the same prepared data.

7. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the major patterns and relationships in the dataset.

The analysis focused on:

- Overall sales performance
- Year-wise sales trends
- Regional sales
- Category performance
- Customer segment performance
- Product performance
- Shipping performance

7.1 Overall Sales Performance

The final Power BI dashboard shows total sales of approximately:

2.26 million

This indicates the overall revenue represented by the analyzed dataset.

The dashboard also reports:

- **Total Orders:** 4,922
- **Total Customers:** 793
- **Total Products:** 1,861

These KPIs provide a high-level overview of the e-commerce business.

7.2 Sales Trend by Year

The sales trend was analyzed across the available years.

The analysis shows that sales decreased slightly from 2015 to 2016. However, sales increased significantly during 2017 and reached the highest level in 2018.

Approximate yearly sales shown in the dashboard are:

Year	Approximate Sales
2015	0.48M
2016	0.46M
2017	0.60M
2018	0.72M

Finding

The increasing sales trend after 2016 indicates strong improvement in business performance during the later years.

8. SALES ANALYSIS BY REGION

Regional performance was analyzed to identify geographical areas generating the highest sales.

The dashboard shows the following approximate performance:

Region	Approximate Sales
West	0.71M
East	0.67M
Central	0.49M
South	0.39M

Finding

The **West region** has the highest sales performance, followed by the **East region**.

The **South region** has the lowest sales among the four regions.

This indicates that regional performance varies considerably and may require region-specific strategies.

9. SALES ANALYSIS BY CATEGORY

Sales were analyzed across the major product categories.

The dashboard contains three major categories:

- Technology
- Furniture
- Office Supplies

Approximate sales performance is:

Category	Approximate Sales

Technology	0.83M
Furniture	0.72M
Office Supplies	0.70M

Finding

Technology is the highest-performing category.

Furniture and Office Supplies also contribute significantly to total sales but remain below Technology.

This suggests that Technology products are an important contributor to overall revenue.

10. SALES ANALYSIS BY CUSTOMER SEGMENT

Customer segments were analyzed to understand which group contributes the most revenue.

The dashboard identifies three customer segments:

- Consumer
- Corporate
- Home Office

The results are approximately:

Segment	Sales	Contribution
Consumer	1.15M	50.76%
Corporate	0.69M	30.44%
Home Office	0.42M	18.79%

Finding

The **Consumer segment** is the largest contributor, accounting for approximately **50.76%** of total sales.

Corporate customers contribute approximately **30.44%**, while Home Office customers contribute approximately **18.79%**.

Therefore, consumer-focused strategies are important for maintaining overall sales performance.

11. TOP 10 PRODUCTS

The project also analyzes the top-performing products based on sales.

The Power BI dashboard identifies the top 10 products by total sales.

Among the highest-performing products are products such as:

- Canon imageCLASS 2200
- Fellowes PB500 Electric Punch
- Cisco TelePresence System
- HON 5400 Series products
- GBC DocuBind products
- GBC Ibimaster products

Finding

The analysis shows that a relatively small group of products contributes significant sales revenue.

The **Canon imageCLASS 2200** is the highest-selling product among the products displayed in the Top 10 analysis.

This information can help businesses prioritize inventory and marketing efforts around high-performing products.

12. SHIPPING PERFORMANCE ANALYSIS

Average shipping time was analyzed based on shipping mode.

The dashboard shows approximately:

Ship Mode	Average Shipping Days
Standard Class	5 days
Second Class	3 days
First Class	2 days
Same Day	Very low / same-day delivery

Finding

Standard Class has the highest average shipping duration, while Same Day has the lowest delivery time.

This indicates a significant difference in delivery performance between shipping methods.

Shipping efficiency is therefore an important operational factor that should be monitored.

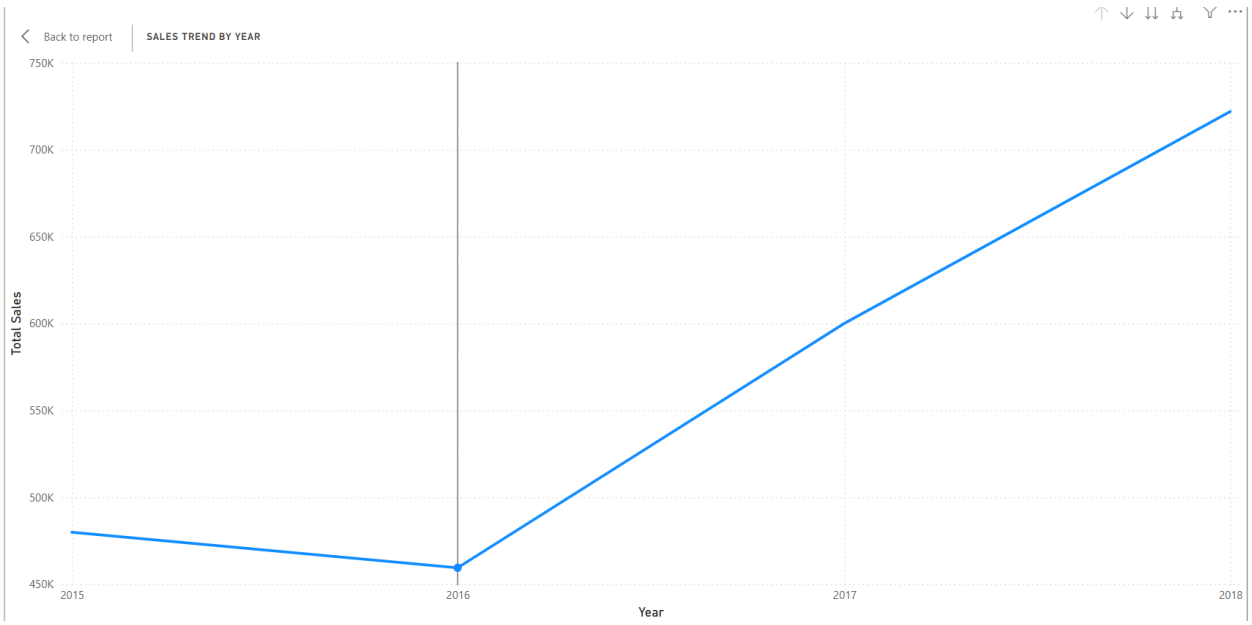
13. DATA VISUALIZATIONS

Several visualizations were created to communicate the important findings clearly.

The major visualizations include:

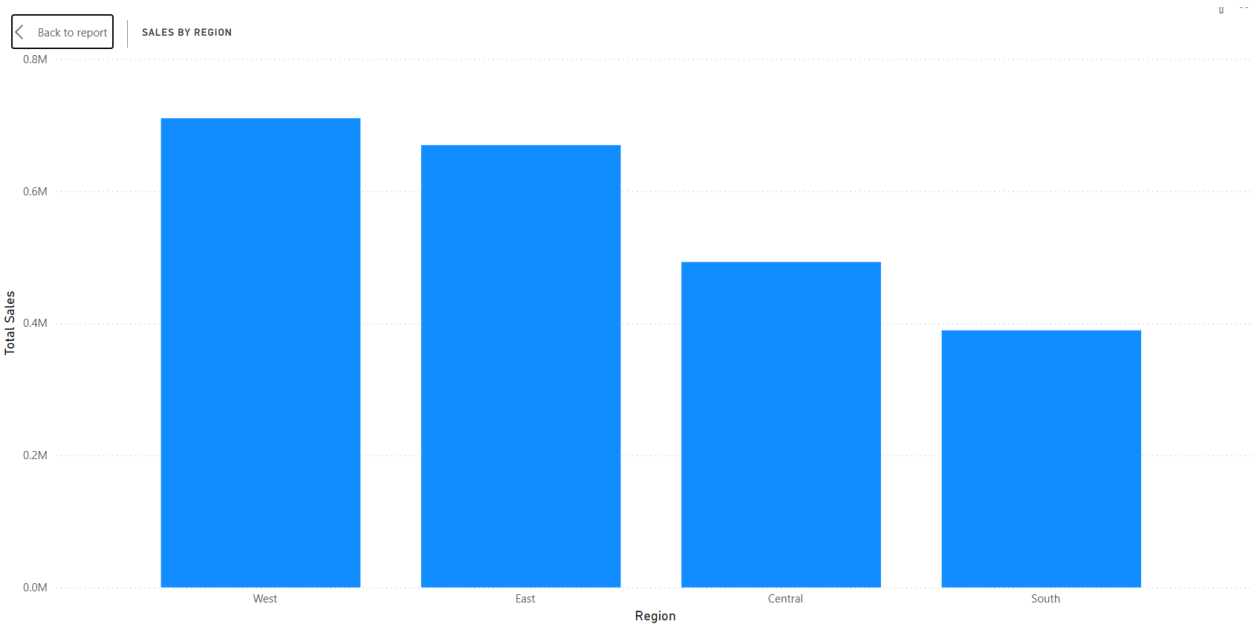
1. Sales Trend by Year

A line chart was used to show the change in sales over time.



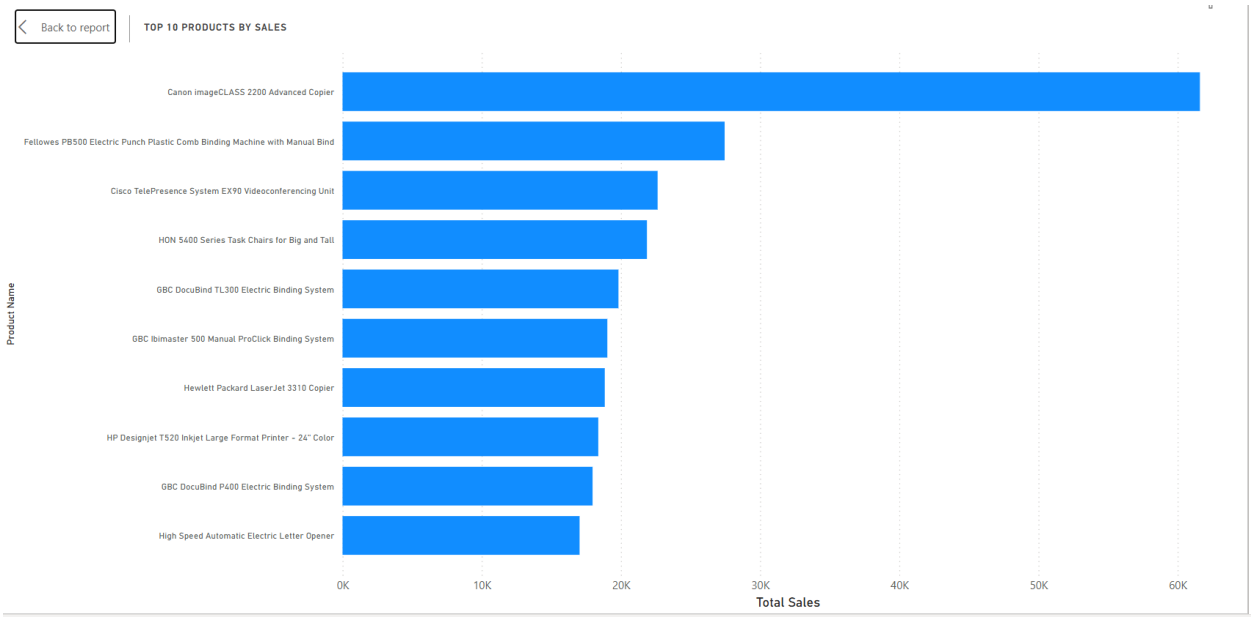
2. Sales by Region

A column chart was used to compare sales performance across regions.



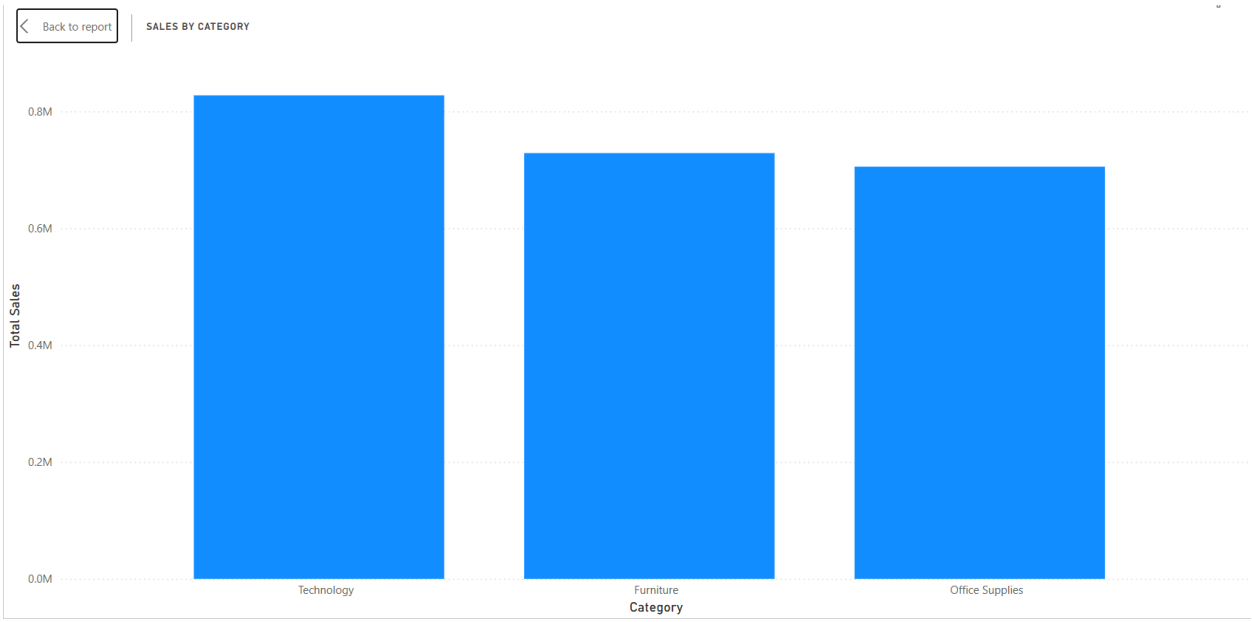
3. Top 10 Products by Sales

A horizontal bar chart was used to identify the highest-selling products.



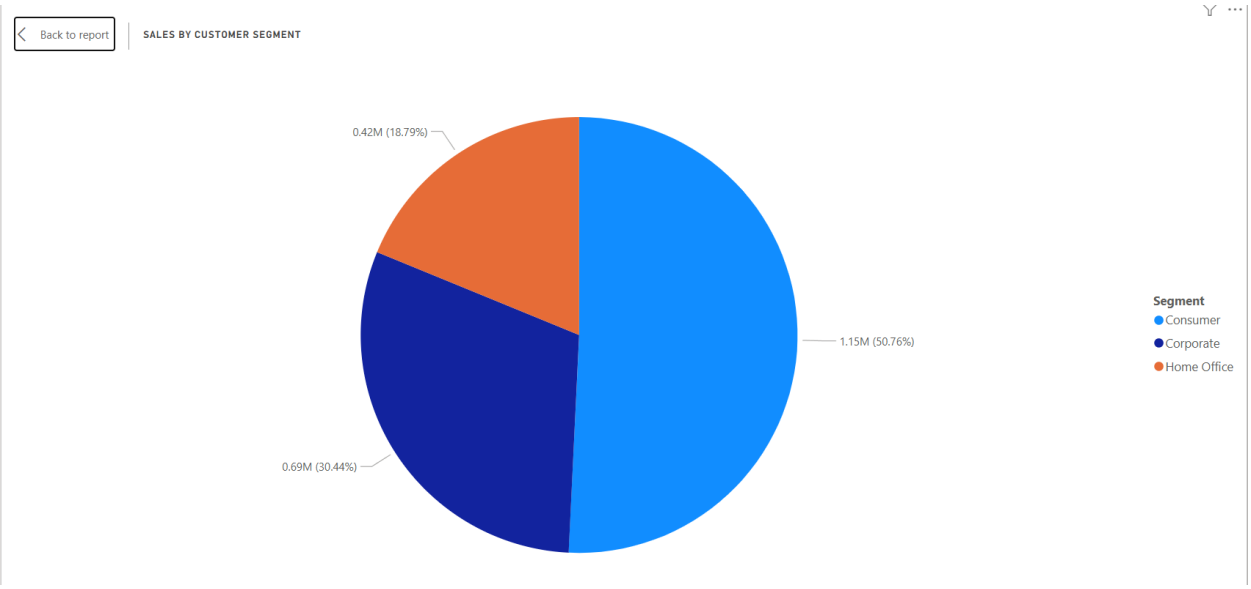
4. Sales by Category

A column chart was used to compare the performance of different product categories.



5. Sales by Customer Segment

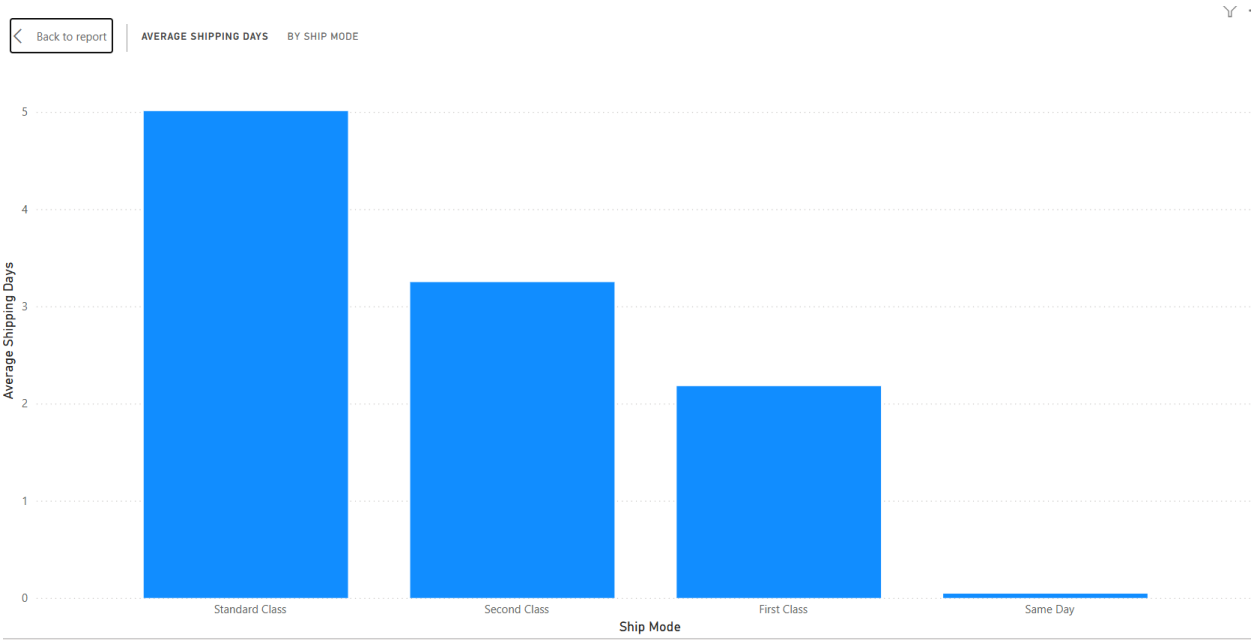
A donut chart was used to show the contribution of different customer segments.



6. Average Shipping Days by Ship Mode

A column chart was used to compare average shipping duration across shipping methods.

These visualizations make the dataset easier to understand and help identify important trends and comparisons.



14. POWER BI DASHBOARD

An interactive Power BI dashboard was developed as the final business intelligence component of the project.

Dashboard Title

E-COMMERCE SALES ANALYTICS

KPI Cards

The dashboard contains four major KPI cards:

- Total Sales: **2.26M**
- Total Orders: **4,922**
- Total Customers: **793**
- Total Products: **1,861**

Dashboard Visualizations

The dashboard includes:

1. Sales Trend by Year
2. Sales by Region
3. Top 10 Products by Sales
4. Sales by Category
5. Sales by Customer Segment
6. Average Shipping Days by Ship Mode

Interactive Filters

The dashboard contains the following slicers:

- Year
- Region
- Category

- Segment
- Ship Mode

These filters allow users to interactively explore the dataset and analyze specific parts of the business.

Dashboard Purpose

The main purpose of the Power BI dashboard is to provide a single interactive view of e-commerce business performance.

It allows users to quickly identify:

- Revenue performance
- Regional performance
- Product performance
- Category performance
- Customer segment contribution
- Shipping efficiency



15. KEY BUSINESS INSIGHTS

Based on the complete analysis and Power BI dashboard, the following major business insights were identified.

Insight 1: West is the highest-performing region

The West region generates approximately **0.71M** in sales, making it the strongest region in the dataset.

The East region follows with approximately **0.67M**, while the South region records the lowest sales at approximately **0.39M**.

This indicates that regional performance is uneven.

Insight 2: Technology is the leading category

Technology generates approximately **0.83M** in sales, making it the highest-performing product category.

Furniture contributes approximately **0.72M**, while Office Supplies contributes approximately **0.70M**.

Technology therefore represents an important revenue-generating category.

Insight 3: Consumers contribute the largest share of sales

The Consumer segment contributes approximately **1.15M**, representing **50.76%** of the sales shown in the dashboard.

Corporate contributes approximately **30.44%**, while Home Office contributes approximately **18.79%**.

This shows that Consumer customers are the most important customer segment in terms of sales contribution.

Insight 4: Sales improved strongly after 2016

The sales trend shows a slight decline from 2015 to 2016, followed by substantial growth in 2017 and 2018.

Sales reached approximately **0.72M in 2018**, the highest annual level shown in the dashboard.

This indicates strong growth in the later years.

Insight 5: Shipping time varies significantly by shipping mode

Standard Class has an average shipping duration of approximately **5 days**, while Second Class takes approximately **3 days** and First Class approximately **2 days**.

Same Day provides the fastest delivery.

This suggests that shipping mode has a significant impact on delivery time and should be considered when evaluating customer service and logistics performance.

16. BUSINESS RECOMMENDATIONS

Based on the findings, the following recommendations are proposed.

Recommendation 1: Strengthen regional sales strategies

The company should continue investing in the high-performing West and East regions while investigating the reasons behind lower sales in the South region.

Possible actions include:

- Region-specific marketing campaigns
- Localized promotions
- Customer segmentation by region
- Improved product availability
- Analysis of regional customer preferences

This can help improve performance in weaker regions while maintaining growth in stronger markets.

Recommendation 2: Prioritize high-performing products and categories

Technology is the strongest category, and the Top 10 Products analysis identifies products that generate significant sales.

The company should:

- Maintain adequate inventory for high-performing products.
- Promote popular products through targeted campaigns.
- Analyze customer demand for high-selling products.
- Avoid stock shortages for high-revenue products.

This can help maximize revenue from products with strong market demand.

Recommendation 3: Improve shipping efficiency

Standard Class has the highest average shipping time.

The company should evaluate:

- Warehouse locations
- Inventory positioning
- Delivery routes

- Logistics partners
- Order processing time

Improving logistics efficiency could reduce delivery times and improve customer satisfaction.

Recommendation 4: Focus on the Consumer segment

Since Consumer customers contribute approximately **50.76%** of sales, the company should develop targeted strategies for this segment.

Possible strategies include:

- Personalized offers
- Loyalty programmes
- Promotional campaigns
- Product recommendations
- Repeat-purchase incentives

These approaches can help maintain and increase consumer revenue.

17. PROJECT STRUCTURE

The project repository is organized as follows:

```
Ecommerce-Sales-Analytics/
│
├── data/
│   └── Dataset files
│
├── notebooks/
│   └── ECommerce_Analytics.ipynb
│
├── powerbi/
│   └── ECommerce_Sales_Analytics.pbix
```

```
│
├── presentation/
│   └── ECommerce_Sales_Analytics.pptx
│
├── report/
│   └── ECommerce_Sales_Analytics_Report.pdf
│
├── visualizations/
│   ├── Sales Trend
│   ├── Sales by Region
│   ├── Sales by Category
│   ├── Top 10 Products
│   ├── Customer Segment
│   ├── Shipping Analysis
│   └── Power BI Dashboard
│
└── README.md
```

This structure makes the project organized and easy to understand.

18. GIT AND GITHUB WORKFLOW

Git and GitHub were used to maintain version control and document the development process.

The basic Git workflow followed during the project included:

Repository Creation

A GitHub repository was created for the project:

Ecommerce-Sales-Analytics

Repository Cloning

The repository was cloned to the local system.

Adding Files

Project files were added using:

```
git add .
```

Creating Commits

Meaningful commits were created during different stages of the project.

Example:

```
git commit -m "Initial project structure"
```

Additional commits were created for project analysis and dashboard development.

Pushing Changes

The changes were pushed to GitHub using:

```
git push
```

Pulling Changes

The repository can also be synchronized using:

```
git pull
```

GitHub provides a centralized location for the project files and documentation.

19. CHALLENGES FACED

During the project, several challenges were encountered.

Data Preparation

Understanding the structure of the dataset and preparing the data for analysis required careful inspection.

Data Visualization

Selecting appropriate charts for different analytical questions was important to ensure that the visualizations communicated information clearly.

Dashboard Design

Creating a Power BI dashboard required arranging KPI cards, charts, and slicers in a clear and organized layout.

Git and GitHub

Learning the Git workflow and resolving repository-related issues helped improve understanding of version control.

These challenges provided practical experience in the complete data analytics workflow.

20. LEARNING OUTCOMES

Through this project, the following skills were developed:

- Data cleaning
- Data preprocessing
- Exploratory Data Analysis
- Descriptive statistics
- Data visualization
- Business intelligence
- Power BI dashboard development
- KPI development
- Interactive filtering
- Business insight generation
- Data-driven recommendation development
- Git basics
- GitHub repository management
- Project documentation

- Data analytics presentation

The project helped connect theoretical concepts with a practical real-world analytics problem.

21. CONCLUSION

The E-Commerce Sales Analytics project successfully demonstrates an end-to-end data analytics workflow.

The project started with dataset inspection and preparation, followed by exploratory data analysis and visualization. The cleaned data was then used to develop an interactive Power BI dashboard containing important KPIs, charts, and filters.

The analysis identified several important findings. The West region was the highest-performing region, Technology was the strongest product category, and Consumer customers contributed the largest share of sales. The analysis also showed significant sales growth in the later years and differences in shipping performance across shipping modes.

The findings provide useful information for improving regional strategies, prioritizing high-performing products, targeting important customer segments, and improving logistics efficiency.

Overall, the project demonstrates how data analytics and business intelligence tools can transform raw e-commerce data into meaningful insights that support data-driven decision-making.

22. FINAL PROJECT SUMMARY

Component	Status
Dataset Selection	Completed
Data Preparation	Completed
Data Cleaning	Completed

Exploratory Data Analysis	Completed
Data Visualization	Completed
Power BI Dashboard	Completed
Business Insights	Completed
Recommendations	Completed
Git/GitHub	Completed / In Progress
Project Report	Completed
Presentation	To be prepared

23. PROJECT REPOSITORY

GitHub Repository:

https://github.com/anuskagupta123/Ecommerce-Sales-Analytics

24. FINAL REMARK

This project demonstrates the practical application of data analytics concepts to a real-world e-commerce dataset and provides a complete workflow from raw data preparation to interactive business intelligence and decision-making.